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24 January 1984
SUPERSEDING
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24 December 1974

MILITARY SPECIFICATION

MELTER, ASPHALT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an asphalt melter for removing asphalt from 55-gallon drums and heating the asphalt to a temperature of 285° F at a rate of 750 gallons per hour (gph).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the Solicitation> form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-C-271	Chains and Attachments, Welded and Weldless.
PPP-P-40	Packaging and Packing of Hand Tools.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Box, Shipping, Fiberboard.

MILITARY

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	Preservation, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: DRDME-DS, Fort Belvoir, VA 22060 Belvoir Research and Development Center, ATTN: Improvement Proposal (DD by using the self-addressed Standardization Document letter. Form 1426) appearing at the end of this document or by

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MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	Treatment and Painting of Material.
MIL-P-14105	- Paint, Heat Resisting (for Steel Surfaces).
MIL-T-22085	- Tape, Adhesive, Preservation and Sealing.

STANDARDS

FEDERAL

FED-STD-595	Colors.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Ildenification Marking of US Military Property.
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-642	Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-l188	Commercial Packaging of Supplies arid Equipment.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS39153	Coupling, Quick Disconnect, 3- and 4- inch, for Construction Equipment.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B31.1 - Code for Pressure Piping.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Structural Steel.
A53 - Welded and Seamless Steel Pipe.
A135 - Electric-Resistance-Melded Steel Pipe.
A659 - Steel, Carbon (0.16 Maximum to 0.25 Maximum, Percent), Hot Rolled Sheet and Strip, Commercial Quality.
C553 - Mineral Fiber Blanket and Felt Insulation (Industrial Type).

(Application for copi should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC.(AWS)

D1.1 - Structural Welding Code, Section 2, Design of Welded Connections.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The asphalt melter shall consist of a melting and storage tank, a dedrummimg tunnel, and all pipe coils, valves, and fittings required to circulate hot oil from an outside source to dedrum and heat grade 85 to 100 penetration asphalt cement at a rate of 750 gph to a temperature of 285°F. The melter shall operate either individually, in pairs, or in trios, in parallel from a single source of hot oil. The melter ready for operation with the doors closed shall be weathertight when tested as specified in 4.5.2.4.

3.2 First article. Unless otherwise specified (see 6.2) the contrator shall furnish two asphalt melters for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce melters that comply with the requirements of this specification. Examination and tests shall be as specified

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in section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first article examination and tests, as specified (see 6.2).

3.3 Production melters. All melters furnished under this specification shall be physically and mechanically identical to the first article melters approved by the Government (see 6.3).

3.4 Material. Material shall be as specified hetein. Material not specified shall b-cd by the contractor and shall be subject to all provisions of this specification (see 6.6).

3. 4.1 Material deterioration and control. The melter shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environments to which the item may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Identification of materials and finishes. The contractor shall ident ify the specific material , material finish or treatment for use with components and sub-components, and shall make information available upon request to the contracting officer or his designated representative.

3.4.4 Structural steel. Structural steel shall conform to ASTM A36.

3.4.5 Sheet steel. Sheet steel shall conform to ASTM A659, UNSG10150, G10200 or G10230, condition and finish to suit the application. All sheet steel gages specified herein shall be United States Standard Gage.

3.4.6 Insulation. Insulation shall be fibrous glass conforming to ASTM C553, type I, class B-4, except the fiber diameter shall not exceed 0.00038 inch.

3.4.7 Pipe. Pipe shall conform to ASTM A53 (API 5LA), type S, grade A or B as applicable, ASTM A53 (API 5LA), type E, grade A or B as applicable, or ASW A135, grade A or B as applicable.

3.4.8 Retaining chain. Retaining chain shall conform to RR-C-271, type II, class 6, 16-1/2 links per foot.

3.4.9 Gaskets and seals. Gaskets or seals, when used, shall be oilproof and rated for operation at 600" F.

3.4.10 Quick-disconnect couplings. Quick-disconnect couplings specified herein shall conform to MS39153.

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3.4.11 Pipe fittings and valves. All valves, caps, couplings, crosses, elbows, reducers, bends, tees, flanges and nipples shall be of steel for 150 psi, 450° F service or malleable iron for 300 psi, 450° F service in accordance with ANSI B31.1. valves used in hot-oil piping shall be of the gate type with rising stem. Each valve shall be labeled to show on-off positions with double ended arrows and the functions served by the valve.

3.4.12 Unions. All unions shall be fabricated from forged steel and shall have stainlees seats.

3.5 Melting and storage tank. The melting and storage tank shall consist of a No. 10 gage minimum sheet steel inner shell, a 2-inch thickness of fibrous glass insulation, and an outer protective sheath made of No. 14 gage minimum sheet steel. Continuous welds shall be used to join the seams on the inner shell. The seams on the outer sheath shall be joined by continuous welding or mechanical means so as to provide west'hertight continuity at all seams. The inner shell shall be leakproof. Frame members shall be of structural steel and shall support a surmounted dedrumming tunnel loaded with twelve 55-gallon drums full of asphalt. The tank shall have an open top in the area directly underlying the tunnel to permit entry of the asphalt to the tank from the drums. The width of the interior of the tank shall be 6 feet +3 inches, and the volume shall be not less than 400 cubic feet. Not less than two layers of 2-inch pipe with welded joints shall be provided at the bottom of the tank to heat the ashphalt to the temperature specified herein. Each layer shall consist of rows of pipe running the full length of the tank and spaced equally across the full width of the tank. The two layers of 2-inch pipe shall be removable so as to provide access to the bottom of the melting and storage tank for cleaning. Unions or companion flanges shall be used at the removable connections. A 2-inch drain, with Chain secured plug, shall be provided in the bottom of the tank. The drain shall be located at the drum discharge end of the melting and storage tank and shall provide complete drainage of the tank when the melter is on a level surface. A 3-inch, asphalt suction pipe with a threaded gate valve shall be provided at the drum entrance end of the melting and storage tank. The discharge end of the asphalt suction pipe shall be provided with the faucet end of a quick-disconnect coupling. The melting and storage tank shall show no evidence of leakage or damage when filled to capacity with water or during dedrumming and melting operations as specified in 4.5.2.2 and 4.5.2.3.

3.6 Dedrumming tunnel. The dedrumming tunnel shall consist of a No. 10 gage minimum sheet steel inner shell, a 2-inch thickness of fibrous glass insulation and an outer protective sheath made of No. 14 gage minimum sheet steel. Continuous welds shall be used to join the seams on the inner shell, The seams on the outer sheath shall be joined by continuous welding or mechanical means so as to provide weathertight continuity at all seams. The tunnel shall have an open bottom and shall be mounted directly over the open section of the melting and storage tank to permit discharge of the asphalt from the drums to the tank. The tunnel shall be constructed so as to provide weatheright continuity at the junction between the mounted tunnel and the melting and storage tank. Frame members shall be of structural steel and shall support the internal piping and twelve 55-gallon drums of asphalt. The tunnel shall be equipped with two

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insulated hinged doors on each end to permit entry and discharge of the drums to and from the drum-heating compartments Doors shall be hinged, shall be provided with latches, and shall be weathertight. Loaded 55-gallon asphalt drums snail slide freely from the loading platform through the entry doors into the dedruming tunnel. The empty drums shall slide freely through the discharge doors into the discharge chute. Three vertical layers of 2-inch pipe with welded joints shall be provided for the tunnel. One layer of pipe shall be placed in the middle to separate t-he two adjacent drum-heating compartments. The other layers shall be placed one on each wall. A horizontal layer of 2-inch, schedule 80 pipe with welded joints shall be provided along the bottom of each drum-heating compartment and shall be spaced so as to support the asphalt arums in the center and on each side of each drum-heating compartment. The tunnel shall be constructed to permit its removal from the melting and storage tank and placing the tunnel in the tank to reduce the total shipping volume to within 110 percent of shipping volume of the melting and storage tank. The tunnel shall be provided with rub strips to prevent damage while being placed in the melting and storage tank and shall be securely anchored when placed in the tank to prevent any damage to the melter during shipment.

3.6.1 Drum hoist. When specified (see 6.2), a manually operated chain hoist shall be provided to place a loaded 55-gallon drum on the loading platform. The hoist load sheaves shall be accurately machined to prevent binding and to pemit easy operation. The hoist shall hold its load indefinitely at any point without help from the operator. The hoist lifting capacity shall be not less than 500 pounds with a chain pull no greater than 75 pounds.

3.6.2 Loading platform. A loading platform for supporting loaded asphalt drums shall be provided. This platform shall extend across the drum entrance end of the asphalt melter and shall support one drum in front of each entrance door. The loading platform shall be provided with an expanded metal surface.

3.0.3 Discharge chute. The asphalt melter shall be provided with a discharge chute for guiding empty drums from the dedrumming tunnel to the ground away from the asphalt melter.

3.7 Hot-oil piping. The hot-oil supply line and return line shall extend to an external location at the drum entrance end of the asphalt melter for hookup to additional hot-oil piping (see 3.10.1), and shall be provided with valves and caps or plugs with retaining chains. All threaded connections shall be NPT. With the exception of connections that are necessary for removal of components, all threaded connections shall be externally brazed, silver-soldered, or welded to prevent leakage. Flanged connections may be used to facilitate component removal. Unions shall be used at all connections which must be disconnected in order to place the dedrumming tunnel into the melting and storage tank for shipment. The melting and storage tank piping shall be arranged in parallel with the dedrumming tunnel piping. Valving shall be provided which will allow the two piping circuits to be operated either individually or in combination at the discretion of the operator.

3.8 Asphalt hose. Unless otherwise specified (see 6.2), each asphalt melter shall be provided with a 10-foot length of 3-inch, high-temperature, flexible,

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galvanized steel hose. Tine hose shall be equipped with an American Standard Pipe thread on one end and the hose end of a quick-disconnect coupling on the other end.

3.9 Thermometer. A dial-type, bi-metal thermometer shall be furnished, shall be provided with a shield to protect the thermometer dial, and shall have a dial range from 50° F LO 450° F. The thermometer shall be provided with a stem which shall be suitable for use in the thermometer well specified in 3.9.1.

3.9.1 Thermometer well. A well for the thermometer (see 3.5) shall be located in the vicinity of the asphalt suction pipe inlet and shall be inclined at an angle above the horizontal.

3.10 Additional piping and fittings.

3.10.1 Additional hot-oil piping. Unless otherwise specified (see 6.2), each asphalt melter shall be provided with additional 2-inch pipes one valve, and all threaded fittings to connect the supply and return lines of two melters in parallel and extend these lines to points shown as "End of Additional Hot-Oil Piping" in figure 1. The required piping shall be based on melters being arranged as shown in figure 1. One valve and all unions and reducers indicated in figure 1 shall be provided with each melter. Piping and fittings shall be arranged so as to incorporate all the valves, unions, and reducers at the locations indicated in figure 1.

3.10.2 Additional asphalt fittings. Unless otherwise specified (see 6.2), each asphalt melter shall be provided with the required 3-inch fittings to connect two asphalt melters in parallel to the suction side of an asphalt distributor. The asphalt hose (see 3.8), shall be utilized between each asphalt melter and these fittings.

3.11 Accessory box. Each asphalt melter shall be provided with an accessory box of sufficient size to hold the additiona] fittings and valves that are removed from the melter for shipment. The box shall be made with a structural steel frame covered with No. 14 gage sheer steel and shall have an overlapping hinged lid which will prevent the entry of rain or dirt. The lid shall be held closed with a hook-type tension fastener. Provisions shall be made to allow the box to drain in its mounted position. The accessory box shall be attached to the melting and storage tank below the drum loading platform.

3.12 Skid. The melting and storage tank (see 3.5) shall be mounted on a structural-steel skid frame to form a self-contained unit. Both ends of the frame shall be reinforced, and each end shall be provided with two towing eyes having an inside diameter opening of not less than 3 inches. The skids shall be curved or turned up at both ends for towing. The skids and skid fastenings shall withstand towing without damage or permanent deformation when the asphalt melter is towed oil a dry level surface at speeds of not less than 2 mph.

3.13 Lifting attachments. Permanently affixed lifting attachments that enable the melter to be lifted in its normal shipping position shall be

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provided. Each attachment shall meet the requirements specified in MIL-STD-209 for type II equipment, class 1 or 3 provisions. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.14 (type II, class 3). The asphalt melter shall also be furnished with lifting attachments which will permit the following operations:

a. Complete asphalt melter to be lifted in its normal operating position.
b. Dedrumming, tunnel to be lifted and placed in melting and storage tank.

The attachments specified in a and b shall be located so that not 1ess than 1-inch clearance is maintained between slings anti all exterior parts while 1 lifting the asphalt melter or dedrumming tunnel, as applicable. Structural members to which the attachments specified in a and b above are fastened and the fastening shall withstand all stresses imposed by the attachment without damage or permanent deformation. Where required, structural members s-nail be reinforced to meet the lifting requirements specified herein.

3.14 Tiedown attachments. Permanently affixed tiedown attachments that permit fastening the melters to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209 for type II equipment, class 2 or 3 provisions, Tiedown attachments may also be used as lifting attachments when such attachments conform to the requirements specified in 3.13 (type II, class 3).

3. 15 Identification marking. The asphalt melter shall be furnished with an identification plate showing the contractor's model number, NSN, serial number, weight, date of manufacture, and date of delivery in accordance with MIL-STD-130. The identification plate shall conform to MIL-P-514, type I, style 1 , composition C, of grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.16 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material shall be provided. in addition, the plate shall show the silhouette of the melter, in transport position indicating the center of gravity and the location and capacity of the slinging and tiedown provisions. The plates shall be attached by screws, bolts, or rivets, in a location where the plate shall be both visible and legible. I

3.17 Instruction plates and labels. Each melter shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the melter. Instruction plates shall conform to MIL-P-514, type III, composition C, or type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible. Labels shall be designed in accordance with MIL-STD-1472, for a viewing distance of 1 to 2 meters.

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3.18 Caution plates. Each asphalt melter shall have a sun resistant caution plate secured to the storage tank warning the operator not to allow the asphalt temperature to exceed 350° F, and to reduce asphalt Temperature to 300° F during standby operation. The words "Warning" or "Caution" shall be red. The caution plates shall conform to MIL-P-514, composition C., or grade A, class 1 material.

3.19 Treatment and painting. The portions of the asphalt melter normally painted shall be cleaned, treated, and painted in accordnce with MIL-T-704, type A when it will not be affected by elevated operating temperature. Heat resistant paint conforming to MIL-P-14105 shall be used at higher operating temperature where type A coating would not be suitable. Color per MIL-P-14105 shall be in accordance with FED-STD-595, lusterless dark green No. 34079 unless otherwise specified.

3.20 Stenciling. The gross weight of each asphalt melter shall be stenciled on each side of the asphalt melter and "Lift" and "Tiedown", as applicable, shall be stenciled adjacent to their respective lifting and tiedown provisions. All stenciling shall be in block-type letters not less than 1 inch high. Registration numbers shall be marked on the asphalt melter in accordance with MIL-STD-642. Color shall be white.

3.21 Safety. With the exception of hot-oil transmission lines, asphalt transmission lines, and unless otherwise specified herein, all exposed parts that are subject to high operating temperatures, or that are energized electrically shall be insulated, enclosed, guarded and labeled to indicate the hazard (for instance "danger hot", "danger high voltage", etc.). Electrical equipment shall be effectively grounded to protect all persons and objects from electrical shock hazards. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Where applicable, all exposed edges and corners shall be rounded to a minimum of .03-inch radius. Sharp corners or edges that are likely to be contacted by personnel shall be suitably protected or rounded to a minimum radius of 1/2 inch.

3.22 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies not normally removed. Covers or plates which must be removed for component repair or replacement shall be equipped with disconnect fasteners. Drain outlets shall be located for accessibility. The use of components requiring special tools for maintenance shall be avoided.

3.23 Workmanship. All parts, components, and assemblies of the asphalt melter including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful extraneous material. All exposed outer edges shall be rounded or chamfered.

3.24 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the asphalt melter to

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conform to the examination and test requirements specified in section 4. Welded joint construction shall conform to AwS D1.1 - Structural Welding Code, section 2, Design of Welded Connections. Unless otherwise specified (see 6.2), the welding process used in fabrication of the asphalt melter shall be at the option of the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility. for insppection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for--the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facitities suitable for the performance of the inspection requirements specified-herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections see forth in the specification where such inspections are deemed necessary to assure supplies and services conform co prescribed requirements.

4. 1.1 Components and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and and tested in accordance with referenced specifications and standards.

4. 1.2 Test support equipment. The contractor shall be responsible for all support equipment required to conduct the test specified herein. The hot-oil heater used to support the melter's operational testing shall not be rated greater than 2,100,000 BTU/hr with an oil circulating range of 100 to 125 gallons per minute at 50 to 75 psi and an outlet temperature no greater than 450° F'.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The asphalt melters shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The asphalt melters shall be tested as specified in 4.5.2.1 through 4.5.2.7. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each asphalt melter shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.2 Component tests. Test each pipe coil before it is installed by capping the ends, subjecting the coil to 100 psi air pressue, and applying a soap solution to each joint. Test each melting and storage tank prior to installing the insulation and outer sheath by filling the tank with water and allowing to stand for 1 hour. Evidence of leaks, or damage to the melting and storage tank or to any of the pipe coils shall be cause for rejection of the component.

4.5 Inspection procedure.

4.5.1 Examination. The asphalt melter shall be examined as specified herein for the following defects:

101. Dimensions not as specified.
202. Fasteners not as specified.
103. Material not as specified or the materials are not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration for the applicable storage and operating environment e
104. The contractor does not have documentation available for identification of material, material finishes or Treatments.
105* Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
106. Assembly not complete or not correct.
107. Components not as specified or missing.
108. Piping arrangement not as specified.
109. Continuity between dedrumming tunnel and melting and storage tank not as specified.
110. Additional hot-oil piping not as specified (when applicable).
111. Thermometer not protected or not located as specified.
112. Drum hoist (when applicable) not as specified.
113. Loading platform not as specified.
114. Discharge chute not as specified.
115. Drain not functional.
116. Accessory box not as specified.
117. Lifting and tiedown attachments not as specified.
118. Identification or other marking, including caution plate, not as specified, missing, or illegible.
119. Treatment and painting not as specified.
120. Stenciling not as specified, missing, or illegible.
121. Workmanship not as specified.
122. Safety devices not as specified.
123. Maintainability not as specified.

4.5.2. 'rests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The melter shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to rest.

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4.5.2.2 Leakage. Fill one melting and storage tank to capacity with water and allow to stand for a period of not less than 12 hours. During the same period, cap and subject all hot-oil piping on the asphalt melter to an air pressure of 100 psi. Evidence of leaks in or damage to either the melting and storage tank or any hot-oil piping shall constitute failure of this test.

4.5.2.3 Performance. Operate the two asphalt melters in parallel circulating hot-oil at a temperature not greater than 450° F from an outside source of hot-oil produced by a hot-oil heater (see 4.1.2). Make the parallel hookup with the pipe and fittings specified in 3.10 and use a distributor as an asphalt transfer pump. Unless otherwise specified (see 3.10.1), the asphalt melters and hot-oil piping shall be arranged as shown in figure 1. Circulate the hot oil through the two asphalt melters with the dedrumming tunnels empty until the oil temperture has stabilized. Then charge the dedrumming tunnel of each asphalt melter with 12 inverted 55-gallon drums (heads removed) full of asphalt cement, grade 85-100 penetration at a stabilized temperature between 65° F and 75° F. After the initial charge has been dedrummed and stabilized at 285° F place a second and third 12-drum charge of 55-gallon drums full of asphalt cement, of the same grade and within the same temperature limits as the initial charge, into the dedrumming tunnel of each asphalt melter. Determine and record the continous time period required to dedrum and melt the asphalt cement and to raise the temperature of the asphalt tamer. : to 285° F for the second and third charges in both asphalt melters. The continuous time period shall begin after the second charge of asphalt cement has been placed in both asphalt melters. The time required for removing the drums from the initial charge and for making the second and third charges shall be included in the recorded continuous time period. Inability of the two asphalt melters to dedrum and melt the asphalt cement and raise the temperature of the asphaly cement (2.640 gallons) to a temperature of 285° F in a continuous time period of 106 minutes or less, asphalt melters or piping not arranged as specified herein, or parallel hookup of the two asphalt melters to the asphalt distributor not as specified herein shall constitute failure of this test.

4. 5.2.4 Simulated rain. After the performance test has been completed, place one of the empty asphalt melters in a position ready for operation and with all doors closed, expose it to a simulated rain for a period of 10 minutes as follows: Eight sprinkler-type nozzles with a 90-degree conical pattern shall be connected to a water supply system which will provide water to each nozzle at a nozzle pressure of not less than 50 psi. Nozzle capacitates at the operating pressure shall not be less than 20 gpm. Two nozzles shall be directed at each longitudinal side of the asphalt melter. These nozzles shall be directed downward at a 45-degree angle from the vertical toward the junction between the dedrumming tunnel and the melting and storage tank. The nozzle directional centerline of one nozzle on each side shall lie in a vertical plane perpem dicular to the side of the dedrumming tunnel located 4 feet from the drum entrance end of the tunnel. (A nozzle directional centerline is defined as an imaginary line through the center of the nozzle head and a point on the asphalt melter at which the nozzle is directed.) The nozzle directional centerline of one nozzle on each side shall lie in a vertical plane perpendicular to the side of the dedrumming tunnel located 4 feet from the discharge end of the tunnel.

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Two nozzles shall be directed downward toward the top of the asphalt melter. One of these nozzles shall be directed toward a point on the longitudinal centerlined of the melter, 2 feet from the drum entrance end of the tunnel, and the other nozzle shall be directed toward a point on the longitudinal centerline of the melter, 2 feet from the discharge end of the tunnel. The nozzle directional centerline of each of these nozzles shall be perpendicular to the top of the asphalt melter. One nozzle shall be directed at each end of the asphalt melter. Each of these nozzles shall be directed downward at a 45-degree angle from the vertical toward the respective end of the asphalt melter at the junction between the dedrumming tunnel and the melting and storage tank. The nozzle directional centerline of each of these nozzles shall lie in a vertical plane perpendicular to the end of the tunnel and located midway between the sides of the tunnel. Each of the eight nozzles shall be located at a distance of 8 feet from the asphalt melter measured along the nozzle directional center-line. Evidence of water leakage on the inside of the asphalt melter shall constitute failure of this test.

4.5.2.5 Towing. Tow one empty asphalt melter in tune operating position over dry, level ground at a speed of not less than 2 mph and in a figure eight pattern laid out to contain a linear distance of not less than 2,000 feet. All addi-tional piping and fittings (see 3.10) shall be carried in the storage cent-ainer during this test. Evidence of weld failure or permanent deformation of or damage to any part of the asphalt melter shall constitute failure of this test.

4.5.2.6 Lifting attachments.

4.5.2.6.1 Test procedure.

a. Lift the melter and hold it in suspension in its normal travel positon using slings that converge not more than 24 feet above the lowest extremity of the melter. Determine the forward angles of application for each provision. Measure the clearance between each sling and the melter. Measure the eye openings and clearance dimensions of the provisions.
b. Restrain the melter by anchoring the main frame and subject each attachment to a force-equal to the load factor specified in MIL-STD-209 times the load it carried in paragraph a above. An acceptable alternate method of test shall be to lift the melter as in a above and add weights to the melter until the required force is obtained. Hold the melter under either of the above loading conditions for not less than 90 seconds.

4.5.2.6.2 Failure criteria. Weld failure, permanent deformation of the melter or any slinging provision, slings converging more than 24 feet above the lowest extremity of the melter, nonconformance to MIL-STD-209, or inability of the dedrumming tunnel to be placed in the melting and storage tank utilizing only those provisions provided for this purpose shall consitute failure of this test.

4.5.2.7 Tiedown provisions.

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4.5. 2.7.1 Test procedure, With the melter anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision, Measure the dimensions of the tiedowns.

4,5.2.7.2 Failure criteria. Weld failure, permanent deformation of the melter or its tiedown provisions, breakage of the previsions, or nonconformance to MIL-STD-209 shall consititue failure if this test.

4.6 Inspection comparison. The Government may select asphalt melters at any time during the contract period and subject these asphalt melteFs to the examination specified in 4.5.1 and to the tests specified in 4.5.2.1 through 4.5.2.5 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site slected by the Government on melters selected-at random from those which have been accepted by the Government and will not include the previously inspected first article. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure Of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

I 4.6.1 Inspection failure. Failure of an inspection comparison melter to meet any requirements specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison melter and shall be cause for refusal by the Government to continue acceptance of production melters until evidence has been provided by the contractor that corrective action has ‘been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on melters previously accepted and produced under the contzact. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all melters accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection pack.

4.7,1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7,1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

124.	Materials, methods, and containers not as specified for level A. Each incorrect material, method, or container shall be considered one defect.	

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125. Tunnel not placed and secured in the tank as specified for level A.
126. Drains in the tank not left open as specified for level A.
127. Oil lines not closed and capped as specified for level A.
128. Surfaces requiring a contact preservative in accordance with referenced document not coated as specified for level A.
129. Drum hoist not placed in box, immobilized, and box sealed as specified for level A,
130. Technical publications not secured to the melter as specified for level A.
131. Maintenance tools not preserved in accordance with the referenced document for level A.
132. Consolidation not as specified for level A.
133. Asphalt hose and hot-oil piping not secured in tunnel as specified for 1evel A.
134. Preservation and packing not in accordance with the referenced document as specified for commercial.
135. Blocking, bracing, and anchoring not as specified for level A.
136. Marking missing, illegible, incorrect, or incomplete for level A or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.l Level A.

5.1.1.1 Disassemble. The dedrumming tunnel shall be removed from the melting and storage tank. The tunnel shall be placed in the melting and storage tank and secured. The drains in the tank shall be left open. Oil lines shall be closed with the caps provided.

5.1.1.2 Unpainted metal surfaces. Unpainted metal surfaces of the asphalt melter, fittings, drum hoist, etc., requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-l preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Drum hoist. When a drum hoist is specified it shall be placed in a close fitting box conforming to PPP-B-636, type CF, class weather-resistant, variety as applicable to the weight. The hoist shall be immobilized within the box to prevent movement. BOX closure and sealing shall be in accordance with the appendix to the box specification as specified for method V.

5.1.1.4 Technical publications. Technical publications when applicable, shall be preserved in accordance with MIL-P-116, method lC-1, or IC-3, and secured to the melter in 2 protected location with tape conforming to MIL-T-22085, type II.

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5.1.1.5 Repair parts. Repair parts when applicable, shall be preserved in accordance with the applicable method(s) of preservation of MIL-P-116.

5. 1.1.6 Maintenance tools. Maintenance tools when applicable, shall be preserved in accordance with the level A preservation and packaging requirements of PPP-P-40.

5.1.1.7 Consolidation. The hot-oil piping and the asphalt hose shall be secured in the dedrumming tunnel. Other loose components shall be consolidated in the accessory box, as space permits, and the box closed and secured. Components that will not fit in the accessory box shall be consolidated together in a close fitting box conforming to PPP-B-601, overseas type, style optional. Strapping of the box shall not be required.

5.1.2 Commercial. Each complete melter shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Each complete melter preserved as specified in 5.1, shall be packed in a crate conforming to MIL-C-104, style A, type and class optional. Blocking, bracing, and anchoring of the contents to the crate base shall be in accordance with the appendix to the crate specification and in accordance with MIL-STD-1186.

5.2.2 Commercial. Each complete melter, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The asphalt melter is intended to be used in removing asphalt from 55-gallon drums and heating and storing the asphalt at a desired pumping temperature.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Time frame required for submission of first article melters (see 3.2).
d. When first article is not required (see 3.2).

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e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2) .
f. When a drum hoist is required (see 3.6.1).
g. When asphalt hose is not required (see 3.8).
h. When additional hot-oil piping is not required (see 3.10.1).
i. When additional asphalt fittings are not required (see 3.10.2).
j. When color is other than specified (see 3.19).
k. When the welding process will not be at the option of the contractor (see 3.24)
l. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. Any changes or deviations of production melters from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish melters conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical. publications, instructional materials, illustrated parts lists, and type of manuals to be furnished with each melter.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair pares and maintenance tools as necessary (including any special tools), and instructions regarding shipment of melters.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specifications (see 3.4).

Custodians:
Army - ME
Navy - YD

User activity:
Army - AT

Preparing activity:
Army - ME

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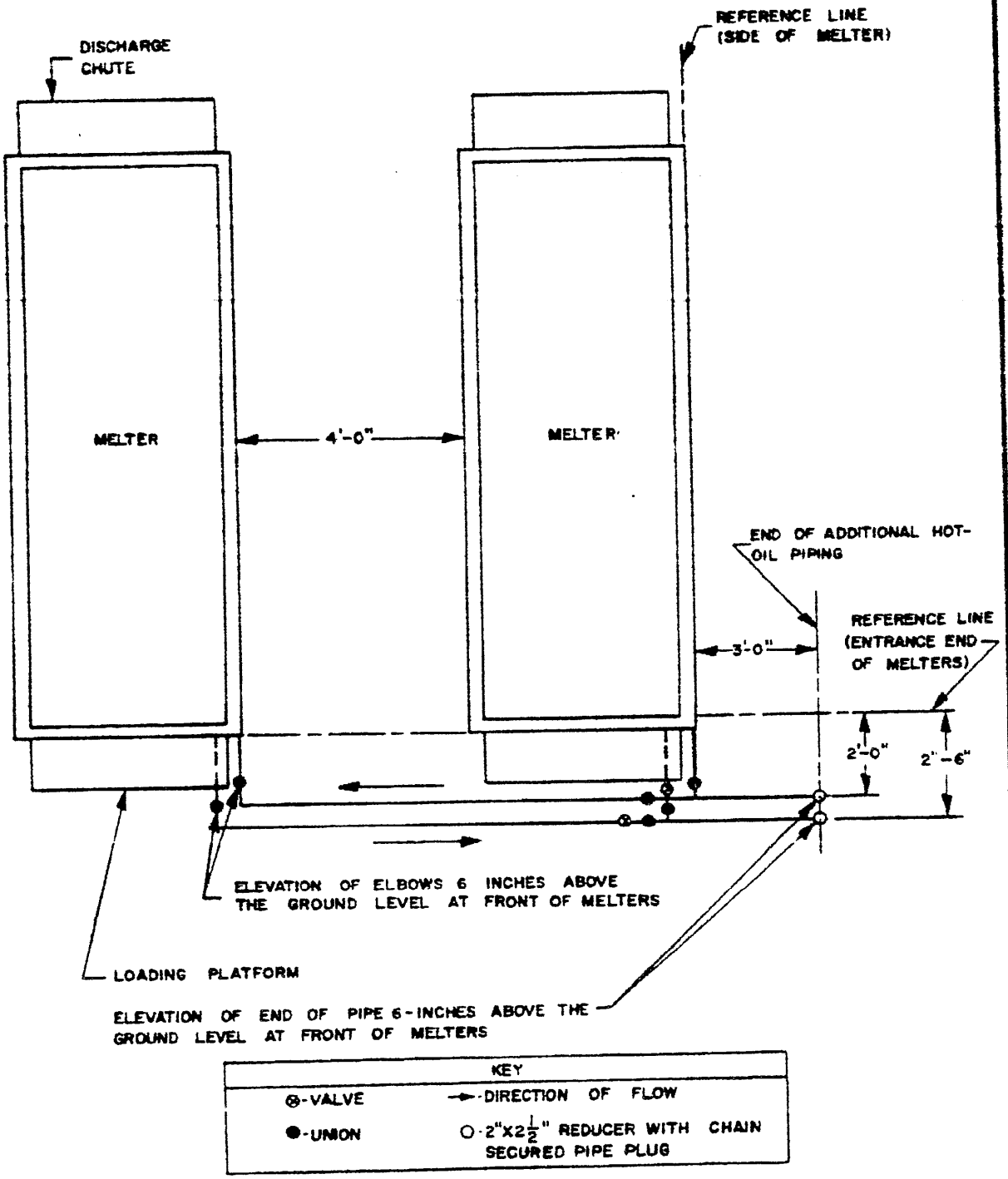


FIGURE 1. ADDITIONAL HOT-OIL PIPING

CX-544

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-52182E	Melter, Asphalt

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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